Post-Fire Vegetation Response in Masticated Chaparral at BCCER

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What I am studying and why

- I am studying how a masticated chaparral site responded to the 2024 Park Fire, with a focus on whether fire mainly reset vegetation structure or also changed plant community composition.

- The Park Fire was a major wildfire, burning about 429,000 acres across Butte and Tehama counties. At BCCER, about 95% of the reserve burned, which created the opportunity to compare the same transects before and after the fire.

Three research questions

- How did woody and herbaceous cover change after fire?
- Did native and non-native species respond differently?
- Did higher fuel consumption predict greater vegetation loss?

18
permanent transects

95%
of reserve burned

2024–2025
pre vs post fire

This matters because mechanically treated chaparral can burn differently from untreated sites, so post-fire recovery needs to be measured, not assumed.

Transect 27: 30 m view facing 0 m

Pre-fire: dense woody shrub structure

Post-fire: open ground and reduced shrub cover

Visual example of structural reset after the Park Fire

Problem and Solution

Problem

After the Park Fire, it was unknown whether the Park Fire caused only structural shrub loss, or also a broader shift in plant community composition and invasion pressure.

This uncertainty spans three dimensions: vegetation structure, species composition, and the role of fire intensity.

Solution

Use paired transect data before and after fire, then test vegetation change at the transect level ,preserving spatial variation and enabling paired statistical inference.

2024
Pre-fire vegetation survey

2024
Park Fire burns through reserve

2025
Post-fire resurvey of same transects

Analysis
Paired tests and regression

Why this approach works

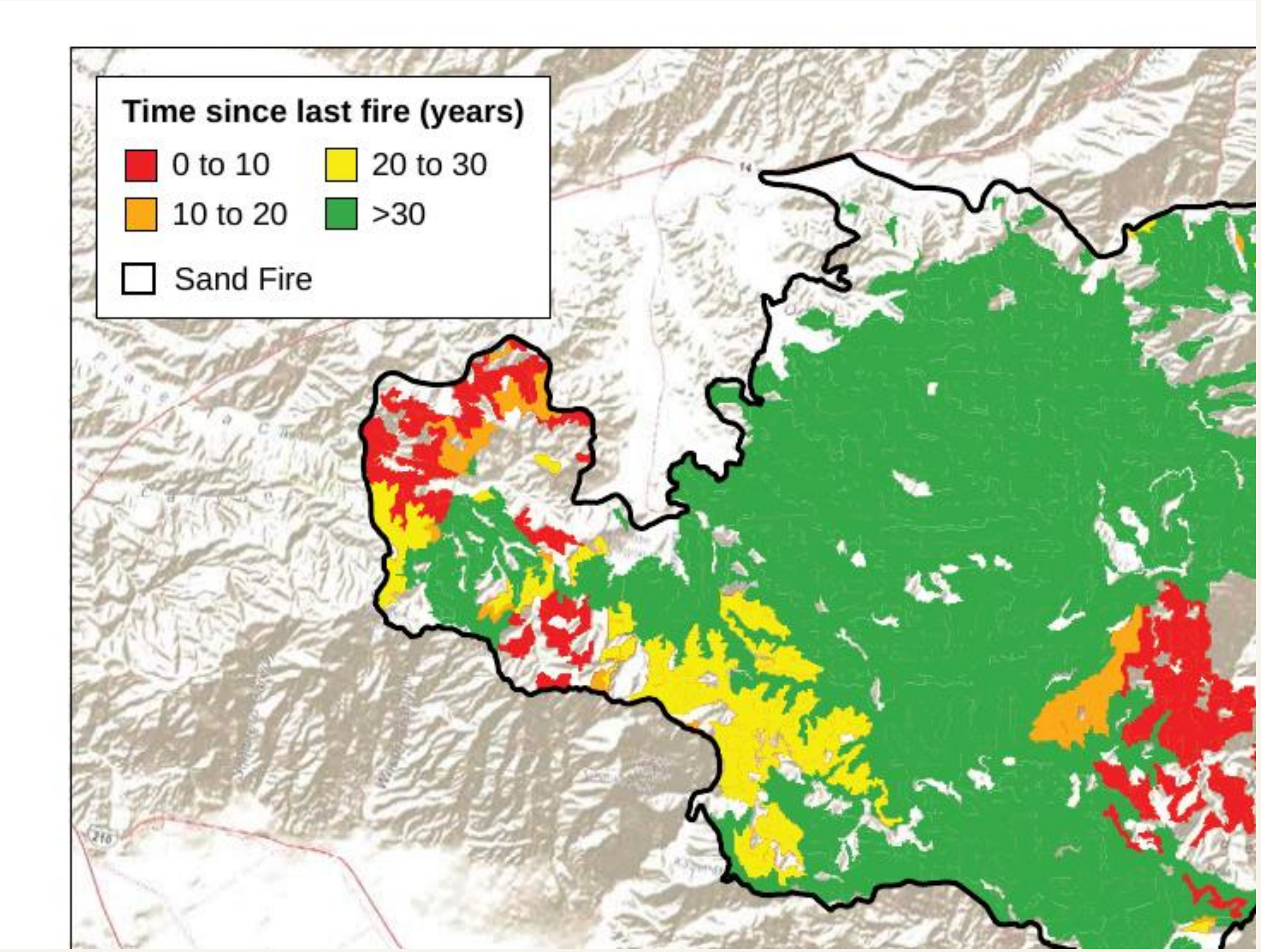
- Keeps spatial differences between transects
- Separates structural loss from compositional change
- Allows fuel consumption to be tested as a driver of shrub loss

Brief literature review

What the literature already shows

- Chaparral is fire-adapted: many native species recover after periodic fire through resprouting or fire-stimulated germination.
- Mastication changes fuel structure: it reduces vertical shrub structure but redistributes biomass into surface fuels, which can affect combustion and soil heating.
- Non-native invasion risk matters: mastication and post-fire openings can favor non-native annual grasses and other invaders.
- Recovery becomes harder when fire returns too soon or when non-native annuals gain a foothold and alter recovery pathways.
- Gap: few studies examine post-wildfire response in masticated California chaparral using paired pre/post transect data and Cal-IPC invasion-risk groups.

Molinari et al., 2021; Potts & Stephens, 2009; Knapp et al., 2009/2011; Brennan et al., 2017; Eliason & Allen, 1997



Example from the literature: short-interval fire can push chaparral toward degradation instead of passive recovery.

Research questions and hypothesis

These questions connect the pre/post vegetation changes to ecological mechanisms and post-fire recovery

Research question	Working hypothesis
How did woody and herbaceous cover change after the Park Fire?	Woody cover would decline more strongly than herbaceous cover because shrubs are more directly exposed to combustion.
Did native and non-native herbaceous species respond differently?	Native cover would be more stable because many chaparral species are adapted to periodic fire.
Does fuel consumed explain vegetation loss?	Fuel consumed would predict woody cover loss more strongly than herbaceous cover change.

Conceptual model: what may be driving response

The key idea: fire seems to have reset structure, while composition responded through additional ecological filters.

Mastication

↓

More surface fuel continuity

↓

Park Fire combustion / fuel consumed

↓

Woody cover loss

Main study question: Is this mostly structural loss, or also a community shift?

Post-fire open space

↓

Light, seed bank, species traits

↓

Variable herbaceous response

↓

Native vs non-native differences

Data Preparation: Raw Field Records

Raw data used:

- Woody species line-intercept records with start and stop positions
- Herbaceous quadrat cover records by species
- Native and non-native species classifications
- Cal-IPC invasive species ratings
- Fuel loading measurements from Brown's planar intercept method

Why this mattered

- Different sheets measured different parts of the ecosystem
- Raw records had to be cleaned and converted before analysis
- The final analysis depended on creating transect-level variables

Data Cleaning and Variable Construction

- Standardized transect IDs and column names
- Corrected reversed woody start/stop entries using absolute segment length
- Removed non-vegetation categories from herbaceous cover:
  - bare ground ,litter ,rock ash ,woody debris ,water
- Tracked incomplete records separately, including MAS19

Cover variables

- Woody cover:
  sum of woody intercept lengths / 30 m × 100
- Herbaceous cover:
  sum species cover within quadrats → average by transect
- Change variable:
  2025 cover – 2024 cover

Species Grouping and Invasion Risk

Why species grouping was needed

- Total herbaceous cover can hide important community differences
- Native and non-native species may respond differently after fire
- Non-native is too broad by itself

Cal-IPC grouping

Non-native species were grouped by invasion concern:

  - High
  - Moderate
  - Limited
  - Not Listed

Why this mattered

- Helped identify which non-native groups drove the overall pattern
- Moderate and Limited groups declined most
- High-rated invasive species did not show a clear decline
- Invasion risk remains important even if total non-native cover dropped

Methods

Paired pre/post design

- This study uses a paired pre-fire/post-fire design at the transect level, so each transect serves as its own baseline.
- Woody cover came from line-intercept segments along 30 m transects.
- Herbaceous cover was aggregated from quadrats after removing non-vegetation classes.
- Species were grouped as native/non-native and by Cal-IPC invasiveness rating.
- Fuel loading was estimated using Brown's planar intercept method.

Statistical approach

- Paired tests compared 2024 vs. 2025 cover at the transect level
- Shapiro-Wilk tested normality of paired differences to choose paired t-test or Wilcoxon signed-rank test
- Linear regression tested whether fuel consumed helped explain cover change

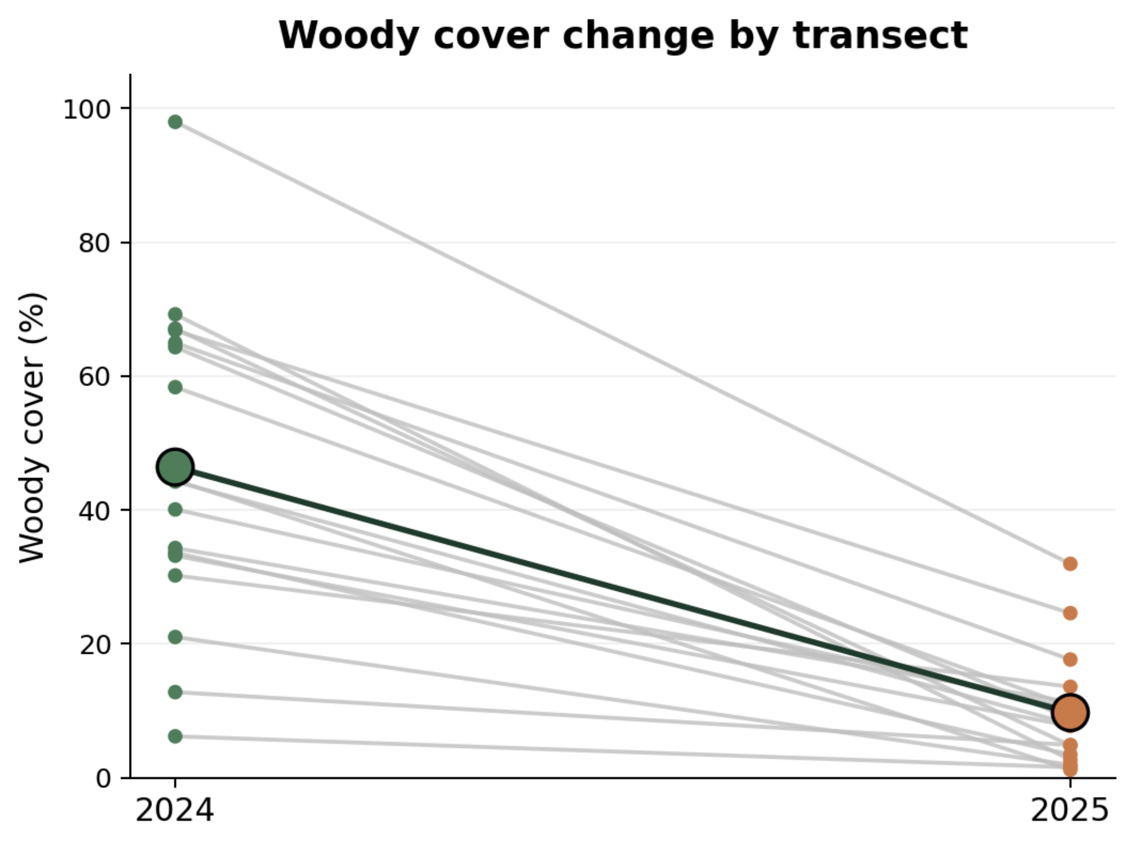
Sample sizes used in results

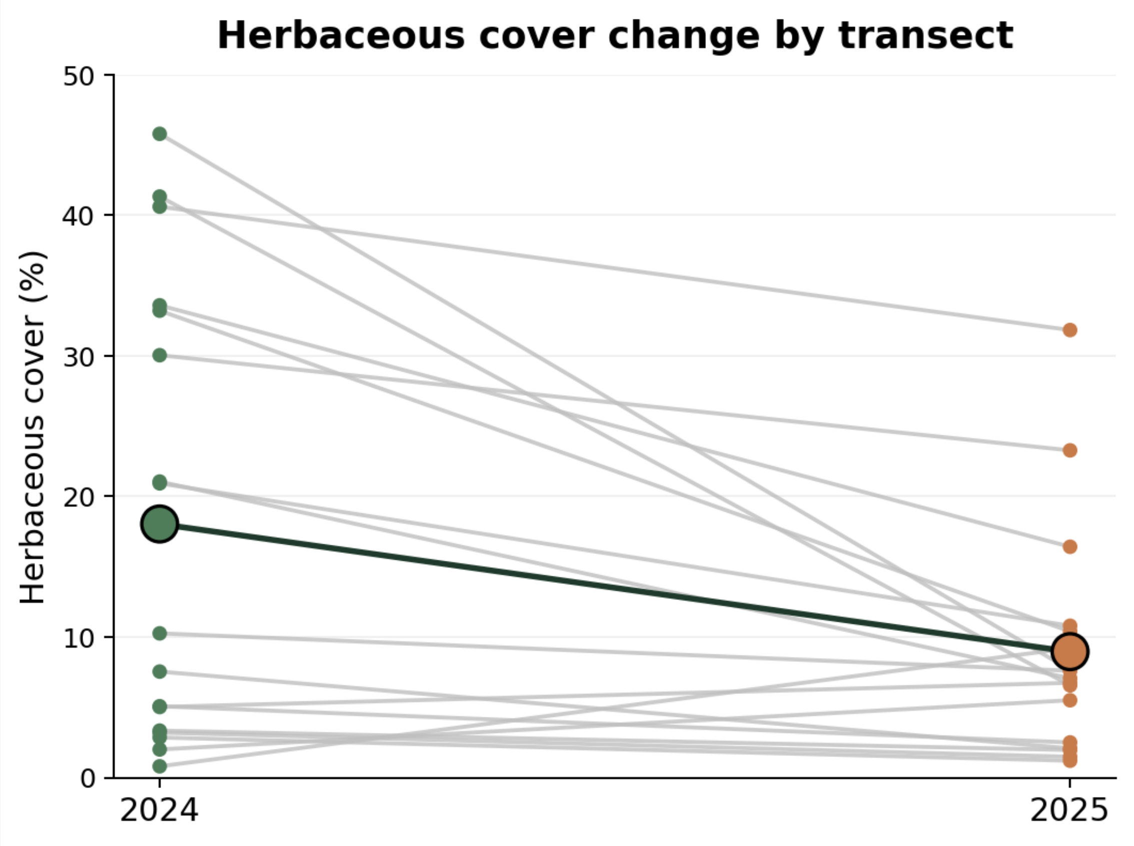
n = 17
woody + fuel analyses

n = 18
native vs non-native

MAS19 had no 2025 woody cover records, so it was excluded from woody and fuel-related cover analyses. It was retained for herbaceous native vs. non-native analyses where valid paired data were available.

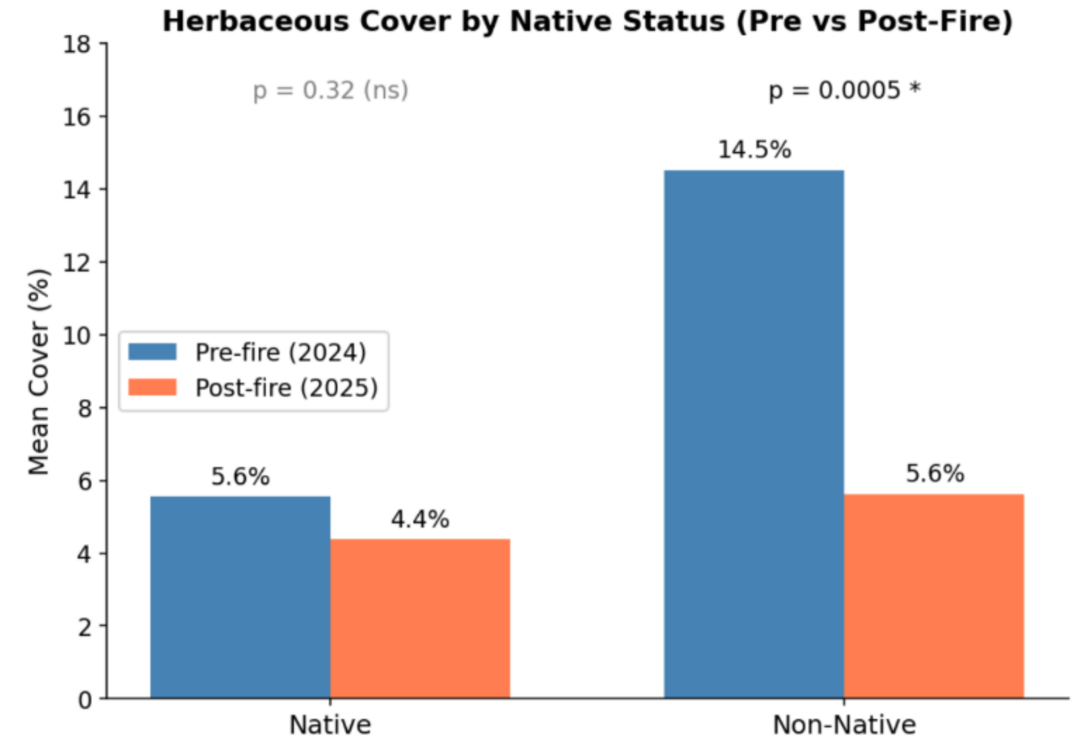
Results: overall cover change

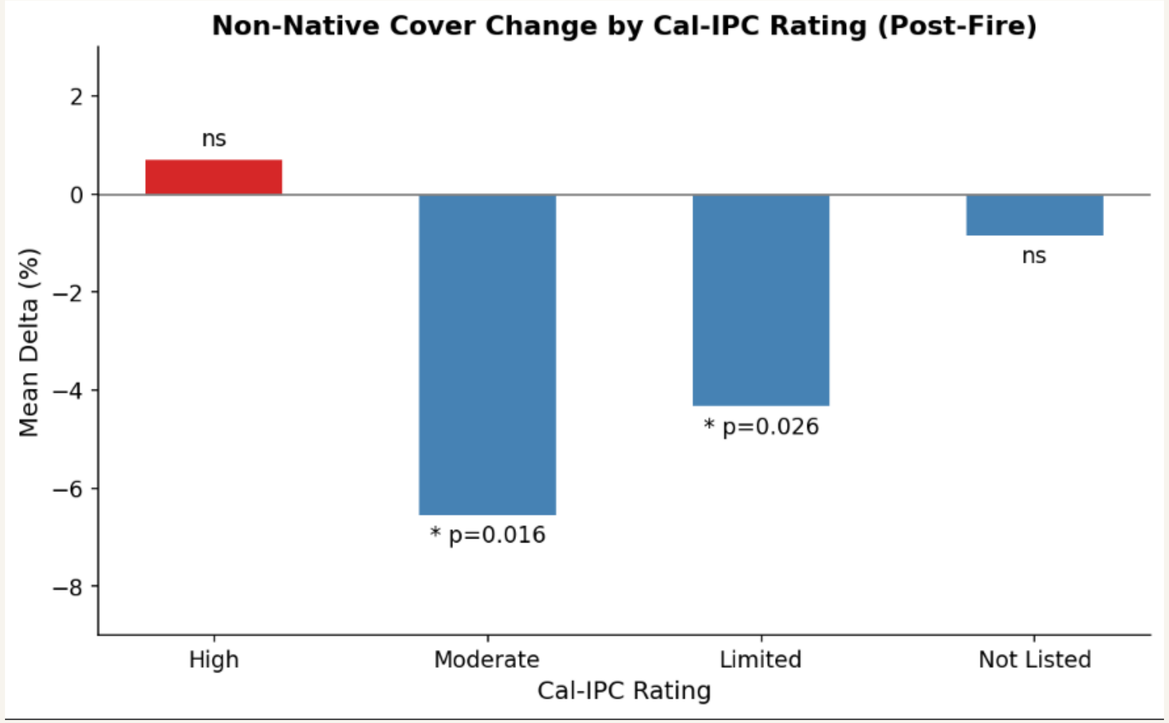




Woody cover: 46.4% → 9.7% (t = −7.78, p < 0.001). Every transect declined. **Herbaceous cover: 18.0% → 9.0%** (Wilcoxon p = 0.0067). Most herbaceous transects declined, but a few increased.

Results: Native status and Invasion risk

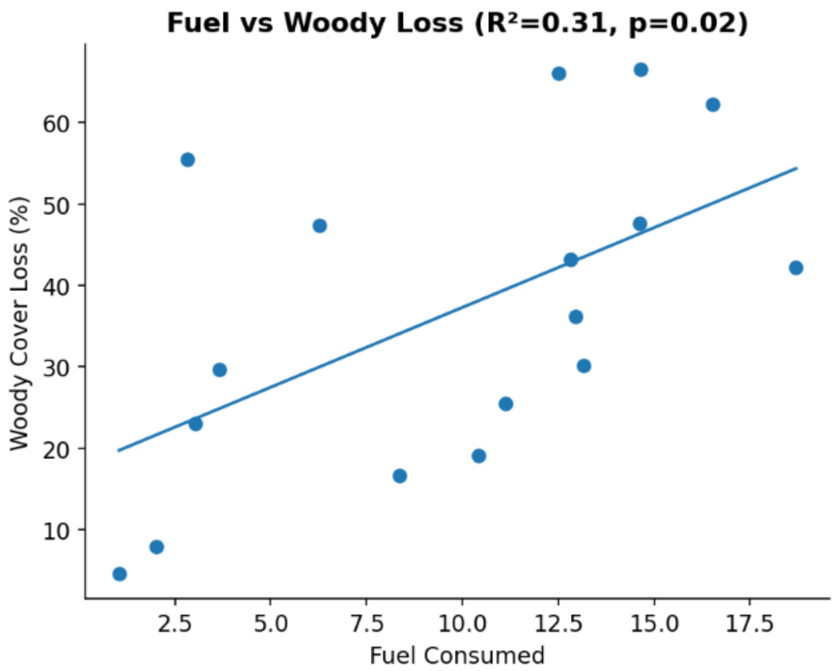


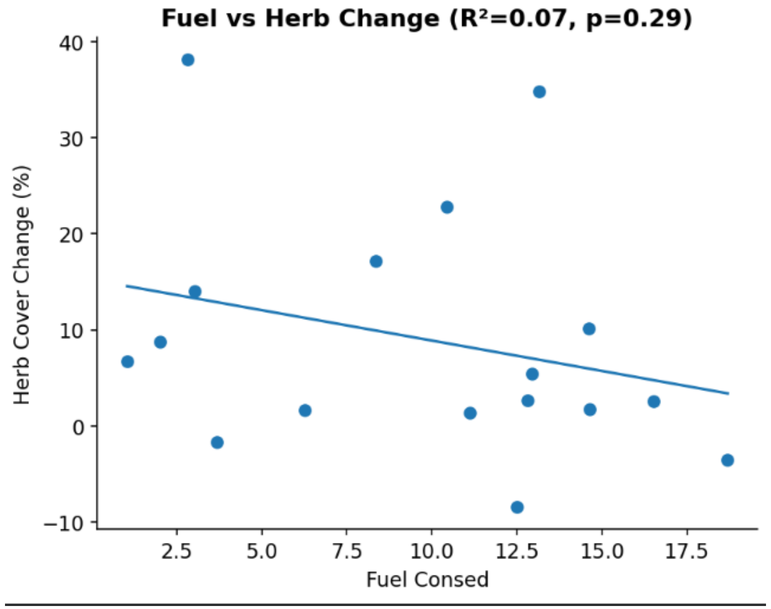


Key points

- Native herbaceous cover changed little: 5.6% → 4.4%, not significant.
- Non-native herbaceous cover declined strongly: 14.5% → 5.6%, $p = 0.0005$.
- The non-native decline was mostly driven by Moderate and Limited Cal-IPC groups.
- High-rated invasive species did not clearly decline, so invasion risk still needs monitoring.

Fuel consumption explained woody loss, not herbaceous change





- Fuel consumed was significantly related to woody cover loss: $R^2 = 0.31$, $p = 0.019$
- Transects with more fuel consumed generally lost more woody cover.
- Fuel consumed did not explain herbaceous cover change: $R^2 = 0.07$, $p = 0.29$
- This suggests woody loss was more tied to combustion, while herbaceous response depended on other factors.

Interpretation: Structure changed more clearly than composition

1. Structure reset

Woody cover collapsed across all paired transects. That makes the fire effect on shrub structure clear, strong, and consistent.

2. Community response was uneven

Herbaceous cover declined overall, but not in the same way everywhere. Some transects decreased strongly, while a few increased.

3. Drivers differ by layer

Fuel consumed explained part of woody loss, but herbaceous change appears to depend more on species traits and local conditions.

Fire created a strong structural reset, but community response was more complex.

Limitations and future work

Limitations

- Only one post-fire time point, so this is short-term response, not full recovery.
- Some subgroups, especially High Cal-IPC species and geophytes, had small sample sizes.
- Fuel consumption is a proxy for fire intensity, not a complete burn-severity measure.
- Slope, aspect, and shrub age were not included and likely explain some remaining variation.

Future work

- Continue monitoring to see whether non-native decline persists or rebounds.
- Add site variables such as slope, aspect, and pre-fire shrub age.
- Bring in burn-severity mapping to strengthen the fire-intensity analysis.
- Track species-level responses, especially High-rated invaders and geophytes.

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THANK YOU